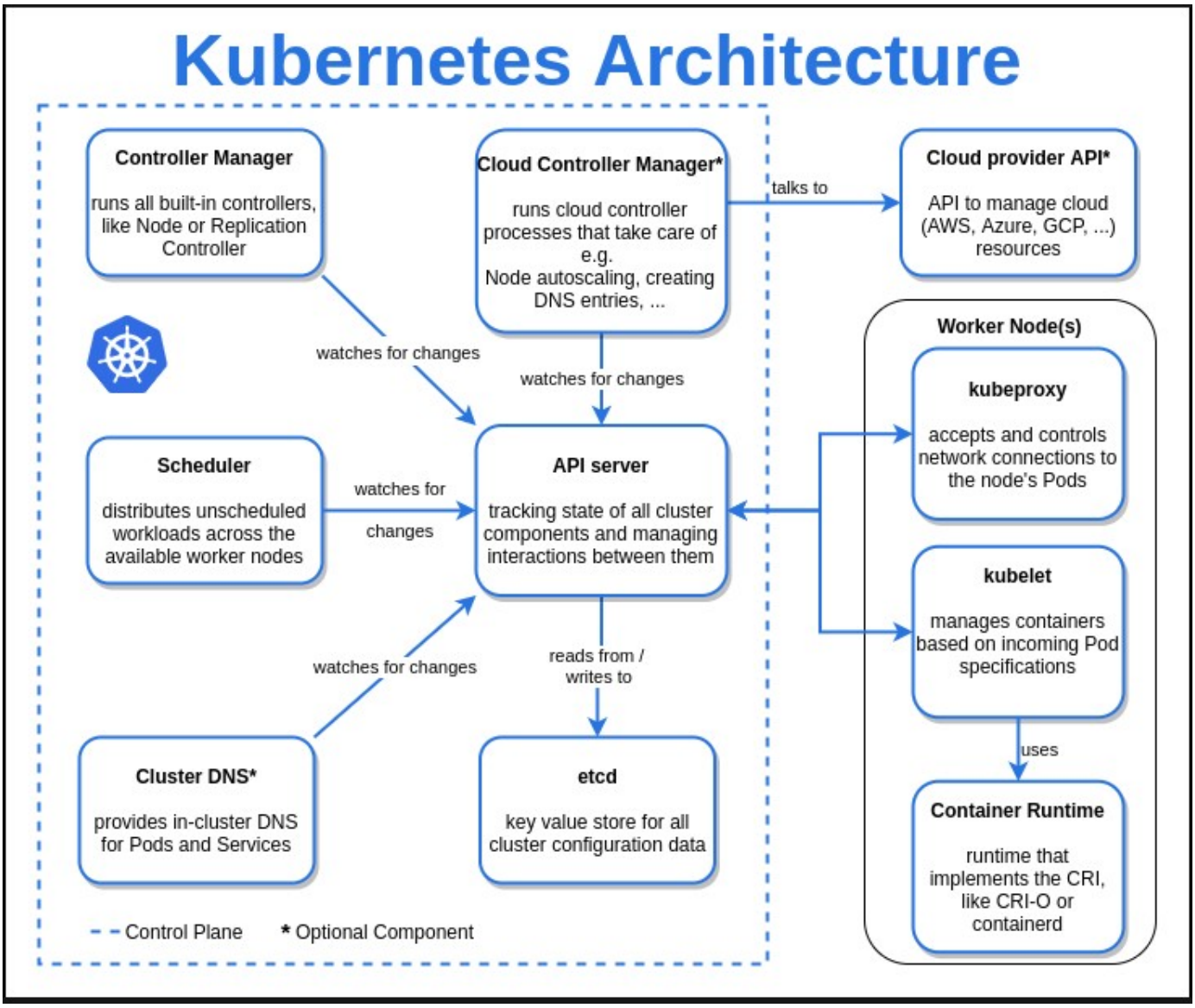
Light

PSScriptAnalyzer passing | Create Release passing | Check kubernetes App failing

LICENSE MIT | FORKS 0 | STARS 0 | Contributors | Issues | LINKEDIN

LEARNING KUBERNETES



Explore the docs »

Main Page - Code Page - Report Bug - Request Feature

Summary

▶ **TABLE OF CONTENT**

About Project

This project aims to help students or professionals to learn the main concepts of kubernetes

(back to top)

Getting Started

This is an example of how you may give instructions on setting up your project locally. To get a local copy up and running follow these simple example steps.

Prerequisites

This is an example of how to list things you need to use the software and how to install them.

- git
- Virtual Box and extension
- Vagrant

Installation

Clone repository

```
git clone https://github.com/marcossilvestrini/learning-kubernetes.git
```

Set ssh keys in security folder

```
# generate ssh key pair for your user access hosts
ssh-keygen -q -t ecdsa -b 521 -N '' -f ~/.ssh/id_ecdsa <<<y >/dev/null 2>&1
cp ~/.ssh/id_ecdsa.pub security/

# generate ssh key pair for rancher
ssh-keygen -q -t ecdsa -b 521 -N '' -f security/rancher-key-ecdsa <<<y >/dev/null 2>
```

Set network

Set network configuration in Vagrantfile. Example:

```
# NETWORK
    ol9_server01.vm.network "public_network", nic_type: "virtio", mac: "080027f3066a
      "Intel(R) I211 Gigabit Network Connection",
      "MediaTek Wi-Fi 6 MT7921 Wireless LAN"
    ]
```

Light

Up kubernetes cluster

```
cd learning-kubernetes/vagrant/linux
vagrant up
```

Usage

Use this repository for get learning about kubernetes exam

(back to top)

Roadmap

- [x] Create repository
- [x] Create github action for automation tasks
- [ ] Install kubernetes cluster

(back to roadmap)

(back to top)

kubernetes Engine work with namespaces(PID,NET,IPC,MNT,UTS) and cgroups.

Kubernetes ports

CONTROL PLANE

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	6443*	Kubernetes API server	All
TCP	Inbound	2379-2380	etcd server client API	kube-apiserver, etcd
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10251	kube-scheduler	Self
TCP	Inbound	10252	kube-controller-manager	Self

WORKERS

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	30000-32767	NodePort	Services All

Install kubernetes

```
foo
```

Kubectl

```
# list nodes
kubectl get nodes
kubectl get nodes -o wide

# delete node
kubectl drain <node_name> --ignore-daemonsets --delete-emptydir-data
kubectl delete node <node_name>
```

(back to install kubernetes)

(back to top)

Contributing

Contributions are what make the open source community such an amazing place to learn, inspire, and create. Any contributions you make are **greatly appreciated**.

If you have a suggestion that would make this better, please fork the repo and create a pull request. You can also simply open an issue with the tag "enhancement". Don't forget to give the project a star! Thanks again!

1. Fork the Project
2. Create your Feature Branch (`git checkout -b feature/AmazingFeature`)
3. Commit your Changes (`git commit -m 'Add some AmazingFeature'`)
4. Push to the Branch (`git push origin feature/AmazingFeature`)
5. Open a Pull Request

License

- This project is licensed under the MIT License * see the LICENSE.md file for details

Contact

Marcos Silvestrini - marcos.silvestrini@gmail.com

Light

Follow @mrsilvestrini

Project Link: https://github.com/marcossilvestrini/learning-kubernetes

(back to top)

Acknowledgments

- CNCF - Cloud Native Computing Foundation
- OCI - Open Container Initiative
- Borg
- kubernetes Website
- Github
- Issues
- CKA Certification
- CKAD Certification
- CKS Certification
- Book Linuxips
- Kind
- Minikube
- k0s
- k3s
- RKE2
- Up RKE2 Cluster HA

(back to top)